

Lab 5: Chat Bot report

CSC 482 - NLP

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Description:

The chatbot is a simple bot that answers on a specific greeting pattern e.g: Hello, how are you, good, great. This chat bot serves 2 main purposes: greeting and answering Q&A. The chatbot is not implemented through complexity. Rather simple primal Q&A system use mostly by pre-define its answer and pre-defined input text word or pattern. The message must follow the pre-defined input for the chatbot to behave accordingly.

Implementation:

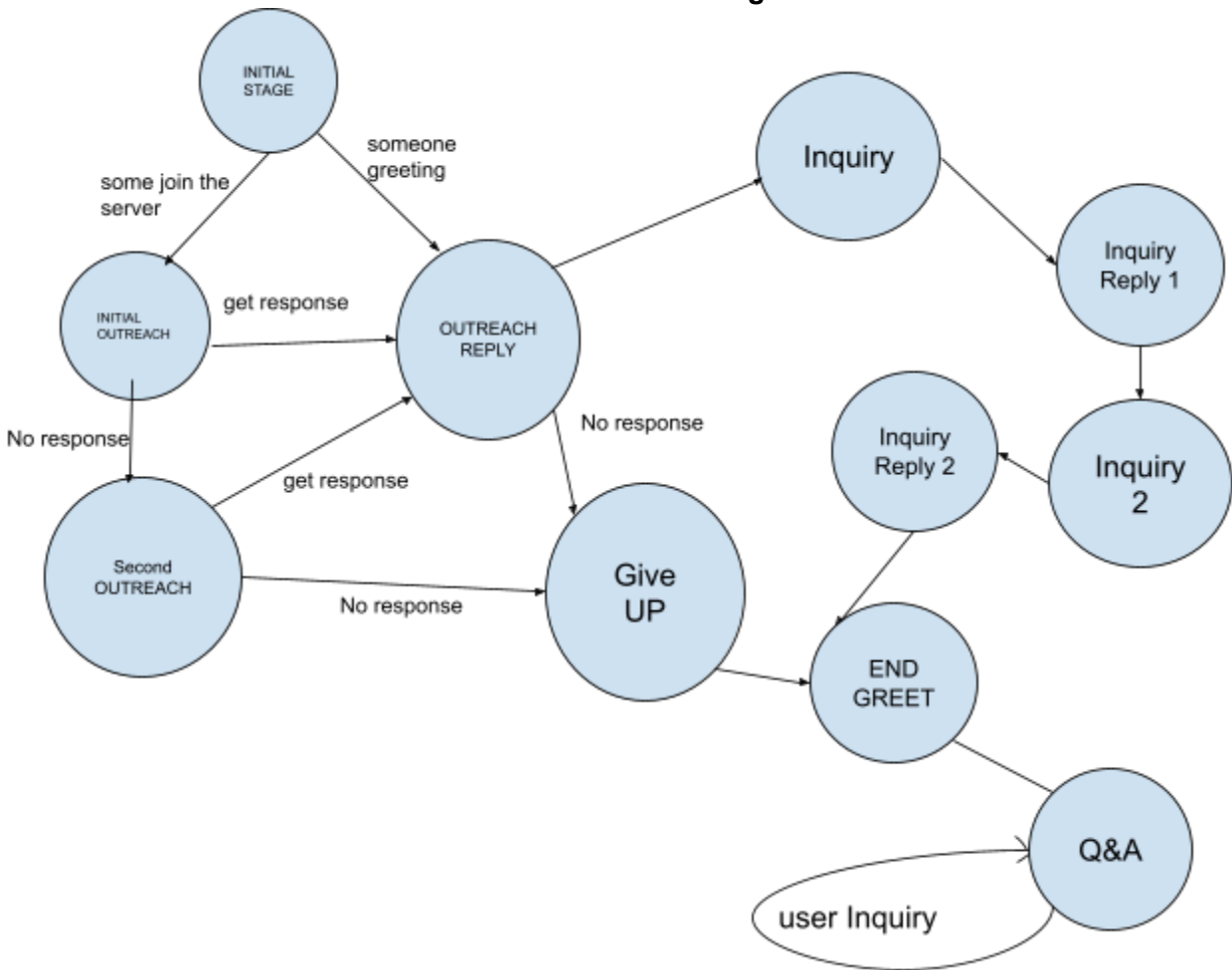
1. Phase I:

- We connect the server by using a socket. We simply use a socket and connect directly by sending a message to the server using starter example (sending bytes of command message of the initiator nickname and channel and then JOIN the server)
- After connecting the server. We keep the program (bot) run on an infinite loop to check to receive a RAW message from the server.
- Through this loop we implemented machine state (Phase II).

2. Phase II:

- **OVER:** We implemented the bot as both as an outreach initiator and an outreach responder
- **STRATEGY:**
 - Each message received from the user the bot transits to the next stage of greeting.
 - After finishing the greeting, the bot enters a QA stage which can answer a question that it is supposed to perform (extra feature).
 - There is a timer at each stage. If there is no response the bot will enter an GIVE_UP stage where it can later answer Q&A.
 - Through the greeting stage, if a person enters a pre-defined message e.g: forget you, screw you. It signals the bot to go to the GIVE_UP stage. The bot then will go straight to the Q & A stage.
- **ROUTINE: HANDLE A COMMAND:**
 - The bot will handle 2 separated kinds of messages: message, command. By using regular expressions the bot will not check if the message is a command (then run a command routine), if not then it will handle the message normally by going through each stage.
- **DATA STRUCTURE:**
 - The program contains multiple bots. Each bot binds 1 on 1 to a user on the server. So the bot has stages of memory for each user it interacts with. By implementing this the bot acts as if it remembers where the conversation was with the user.

PHASE II figure



3. Phase III:

- We decide to implement mapping functionality for the phase 3 of the chat bot
- The chat bot checks regular expressions from the user input. If it valid, (matches "where is <latitude> <longitude>" or "where is the ISS"), the bot replies with the best known location
 - If the user specifies "where is the ISS", the International Space Station's current location is searched using the Open Notify API.
 - Otherwise, the user's specified latitude and longitude are used
 - Address information is then searched using the Open Street Map API
 - If an address city, state, and country are found, those values are recorded

- Else if an address name is found, the address name is recorded
- If an address cannot be found, the IHO World Sea Areas Shapemap from Marineregions.org is queried to find which sea the location is in, and the sea name is recorded
 - If no sea is found and we found the ISS location, we record the latitude and longitude. Otherwise if no sea is found, we record we could not find a location.
- We message the user the best recorded location
- The map Q&A stage loops forever until the bot is killed by command "die" or command "forget".